CSS452 Internet of Things

Homework 3: ESP32 – Web Servers and Clients

1. Do the project in Section 4 "ESP32 as a Local Web Server to Monitor Sensor Data". Take a video to show

    - circuit connection,
    - Arduino code,
    - demonstration:
        - The potentiometer value is shown on the webpage.
        - The potentiometer value on the webpage is updated when we rotated the knob of the potentiometer (you will need to press the refresh button on the web browser).

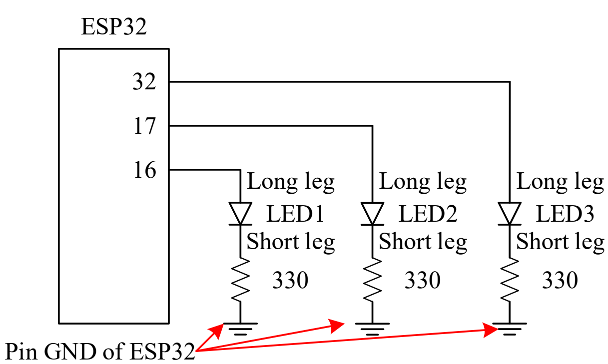
    Name your video as "P1" and submit to the Google Classroom.

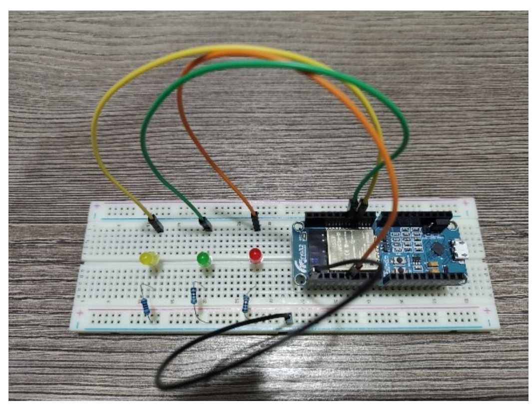
2. Do the project in Section 5.2 "Sending ESP32 Data to the ThingSpeak Cloud". Take a video to show

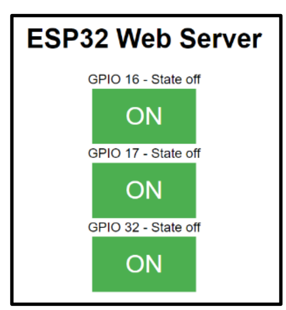
    - circuit connection,
    - Arduino code,
    - demonstration:
        - The potentiometer value and a random number are shown on the ThingSpeak charts.

    Name your video as "P2" and submit to the Google Classroom.

3. Connect three LEDs to the ESP32 as shown below.







Similar to the project in Section 3, write an Arduino code such that the ESP32 will be a webserver and provide a webpage consisting of three buttons. By using a web browser, pressing these buttons will turn on/off these LEDs accordingly (i.e., revise the Arduino code in Section 3).

Upload your code to the ESP32 and take a video to show

- circuit connection,

- Arduino code,
- demonstration:
  - Press the button (GPIO16) on the webpage to turn the LED1 on/off.
  - Press the button (GPIO17) on the webpage to turn the LED2 on/off.
  - Press the button (GPIO32) on the webpage to turn the LED3 on/off.

Name your video as “P3” and submit to the Google Classroom.
